Tshwane University of Technology

We empower people

YEAR: 2023
SEMESTER: 2
ASSESSMENT: Formative Assessment

SUBJECT NAME:	Programming Principles
SUBJECT CODE:	PPB115D/PPG115D/PPBF15D/PPGF15D
QUALIFICATION(S):	DPIF20/DPMC20/DPRS20/DPIT20

PAPER DESCRIPTION: CLOSED BOOK **DURATION:** 3 HOURS **PAPER:** ONLY

SPECIAL REQUIREMENTS

- ☐ **NONE**
- ☐ **NON-PROGRAMMABLE POCKET CALCULATOR**
- ☐ **SCIENTIFIC CALCULATOR**
- ☐ **COMPUTER ANSWER SHEET**
- ☐ **GRAPH PAPER**
- ☐ **DRAWING INSTRUMENTS**

OTHER:

INSTRUCTIONS TO CANDIDATES: ANSWER ALL QUESTIONS

CHECK EACH SECTION OR QUESTION FOR ADDITIONAL INSTRUCTIONS

TOTAL NUMBER OF PAGES INCLUDING COVER PAGE: 10

TOTAL NUMBER OF ANNEXURES: 0

EXAMINER: Miss VM Booi
MODERATOR: Mr M. Liebenberg

FULL MARKS: 90
TOTAL MARKS: 90
STUDENT TOTAL: ___
STUDENT %: ___

INSTRUCTIONS ON HOW TO WRITE THE TEST ON ELECTRONIC CAMPUS (EC):

1. Create a folder on **D:\ Drive** and rename the folder your **student number**.
2. Save all your work in the folder you have created
3. Open a text editor (notepad++ is recommended) or MS Word and save your document in your student number folder.
4. Open the **browser**.
5. Type http://ec.tut.ac.za in the address field of the browser and press enter.
6. Click **Login.**
7. Type in student number for user name and if you are using EC for the first time you must use your student number as your password. (Not applicable any more. Please change to the current situation).
8. Alternatively, you must enter your password. (Same here).
9. Click **Login button**.
10. Click [**Continue**]
11. Select [**Student**]
12. Select [**Tests**]
13. Select [**Webtest 4**]
14. Type or Select **PPB115D/PPG115D/PPBF15D/PPGF15D** into the first text field.
15. Type or Select **20232** into the second text field.
16. Type or Select **T** into the third text field (The "**T**" must be a capital letter).
17. Click "**Next**"
18. Click on the "**Write**" button.
19. Click on "**[Start]**" and you will be in the test to submit your answers.
20. Click on **[Section]** on the test to submit each Question, and click [Go] to submit your answer- Copy your answer and paste it on the space provided.
21. Click **[Test]** to move to the next or previous Questions.
22. Under each question click edit to update your answer.

Question 1 **[5]**
True/False

Instructions
Log on to EC and proceed to the correct test. Select **Section 1** and provide your answers to this question therein.

Questions

NO.	STATEMENT
1.1	Arrays are objects and they are considered as reference types.
1.2	When creating an array all elements in the array are given default values.
1.3	Parallel arrays in Java is when you have two or more arrays in your application.
1.4	Arrays can be passed as parameters to a method either by value or by reference.
1.5	When manipulating arrays, the index of the array must always start at 0.

Question 2 **[10]**

Multiple Choice

Instructions

Log on to EC and proceed to the correct test. Select **Section 2** and provide your answers to this question therein.

Questions

2.1 An array is described as _____.

a. A collection of elements with different types
b. A collection of elements with the same type
c. A resizable data structure
d. A container for storing key-value pairs

2.2 Which of the following can be used as an array subscript?

a. char
b. String
c. double
d. int

2.3 What happens if you try to access an array element with an index that is out of bounds?

a. Causes a runtime error
b. The element value is set to null
c. The element value is set to 0
d. None of the above

2.4 Which of the following is a valid statement for declaring and initializing a one-dimensional array?

a. int[-5] myArray;
b. int[5] myArray= { 1, 2, 3, 4, 5, 6 };
c. int[] myArray= new myArray[];
d. int[] myArray= { 1, 2, 3, 4, 5, 6 };

2.5 Which of the following is an invalid statement to declare and initialize a one-dimensional array?

a. boolean[] QuziResponses= {true, false, true, false, true, true, false};
b. double[] busFare= {50.99,36.99, 75.99};
c. int[] noTickets= { 4; 15; 13; 55;105};
d. String[] zipCode= {“ZA” ,”AUE”, “UAE”};

2.6 Assume x is an integer and an array is declared as follows. Which of the following statements correctly assigns the value 500 to each of the array elements?

double[] salaries = new double[4];

a. for(x = 0; x < 3; ++x) salaries[x] = 500;
b. for(x = 0; x < 4; ++x) salaries[x] = 500;
c. for(x = 1; x < 4; ++x) salaries[x] = 500;
d. for(x = 1; x < 5; ++x) salaries[x] = 500;

2.7 A parallel array is one that _____.

a. holds values that correspond to those in another array
b. holds an even number of values
c. is placed adjacent to another array in code
d. is placed adjacent to another array in memory

2.8 A single array element of a primitive type is passed to a method by _____.

a. value
b. reference
c. address
d. both by value and reference

2.9 When you place array elements in order beginning with the element with the highest value, you are sorting in _____ order.

a. descending
b. ascending
c. demeaning
d. arithmetic

2.10 Bubble sort involves comparing each array element with _____.

a. the corresponding element in a parallel array
b. the adjacent element
c. the arithmetic average
d. the lowest element value

Question 3 **[16]**

Short Questions

Instructions:

Log on to EC and proceed to the correct test. Select **Section 3** and provide your answers to this question therein.

Preparations

- Open Microsoft Word or Notepad and type in your answers therein.
- Question 3.1 to Question 3.2 answer must be typed onto this notepad/word document.
- Submit each question under the slot/section provided on EC.
- Make sure you submit your answers every time when you are done with a question.

Questions

3.1 Write **only** the required statement for the following scenarios using **Java syntax** for each.
You have two parallel arrays called **courseNames** and **courseFees** that will store the course information. The course fees array has been declared and initialized with the given fees. There are 10 courses offered.

3.1.1 Declare the **courseNames** array using the course fees array sizes in your declaration**.** (2)
3.1.2 Write a statement that will increase the fees for the third course in the array with 10%. (2)
3.1.3 Write a statement to display the last records in the two parallel arrays using this format: (2)
Course Name Course fees

3.2 Write **only** the required statement or statements.

Consider the following description:
A company has four different learnership opportunities as indicated in Table 3.1. The learnership positions are stored in an array called **vacancies** in the main method. The program will prompt the user to enter a vacancy to search for in the main method as well using a variable name called **postSearch**. The program must then allow the user to search for vacancies in a method called **findLearnership**. If a match is found the method must return true, otherwise a false must be returned.

Table 3.1

Vacancies
Data Scientists
Java Developer
UX Designer
Project Manager

3.2.1 Write a call statement for the method using an if-statement. (2)
3.2.2 Write a method definition/implementation to search for a vacancy for the above description in Java. (8)

Question 4 **[13]**

String Manipulation using arrays

Instructions

Log on to EC and proceed to the correct test. Select **Section 4** and provide your answers to this question therein.

Preparations

- Create a folder in the **D** drive called **"D:\My Documents\StudentNumber\Question4\"**, where **StudentNumber** is your actual student number.
- Create a new file in **Notepad++** called "**StringManipulation.java**". Open the file and complete the question as instructed below.

Completion time: 30 Minutes

Short Program: String Manipulation

Develop a Java application that will manipulate a String. The program must allow the user to enter a string as input and this must be stored in an array. The application must then manipulate the given String by modifying every 5th character in a String with an asterisk (*). Display the length of the string and the modified string in uppercase.

Sample Output:

```
Enter your string here :I love programming in Java!!
Length of this string =28
I LO*E PR*GRAM*ING *N JA*A!!
```

Figure 4.1

Question 5 **[46]**

Arrays and Static Methods

Instructions

Log on to EC and proceed to the correct test. Select **Section 5** and provide your answers to this question therein.

Preparations

- Create a folder in the **D** drive called **"D:\My Documents\StudentNumber\Question5\"**, where **StudentNumber** is your actual student number.
- Create a new file in **Notepad++** called "**StreamingServicesApplication.java**". Open the file and complete the question as instructed below.

Streaming Service Provider Application

PayTV SA is an organization that monitors usage of pay television, this type of business model depends solely on monthly subscription fees paid by individual customers. The organization likes to view/ monitor the statistics of the current service providers they have and to look at which one is the most watched. Below is the information on the service providers they want to monitor.

Table 5.1: Streaming Service information

serviceProviders	originCountry	subscribers	userRatings
YouTube	United States	0	0
Netflix	United States	0	0
DSTV	South Africa	0	0
ShowMax	South Africa	0	0
PrimeTV	United States	0	0
SABC+	South Africa	0	0

They want you to create an application that will help monitor the streaming services statistics.

To successfully complete the application, follow the UML diagram shown in **Table 5.2,** and the corresponding description of the UML in **Table 5.3**, Sample output is provided in **Figure 5.1** to **Figure 5.3.**

Table 5.2: UML diagram of StreamingServicesApplication class

StreamingServicesApplication
+ *main*(String []): *void* + *populateSubscribers(int[]): void* + *populateRatings(int[], String[]):): void* + *mostWatchedService (int[]): int* + *sortServiceProvider (String[], String[], int[], int[]):void* + *displayStreamingServices(String[], String[], int[], int[]): void*

Table 5.3: Description of StreamingServicesApplication class

	Method name	Description	Marks
5.2.1	**main(String [])**	In this method, do the following: • Declare two parallel arrays named **serviceProviders**, **originCountry**, and initialise them using the information given in **Table 5.1**. • Declare another two parallel arrays named **subscribers** and **userRatings**. • Populate the array called **subscribers** by calling/invoking/using the method called **populateSubcribers()**. • Populate the array called **userRatings** by calling/invoking/using the method **populateRatings().** • Determine the most watched service provider by calling/invoking/using the method called **mostWatchedService().** • Display the most watched service provider with the corresponding number of subscribers as indicated in **Figure 5.2**. • Call/invoke/use the method **sortServiceProvider()**. • Display the sorted array values by calling/invoking/using the method **displayServiceProviders()** to display the service provider details as indicated in **Figure 5.3**.	14
5.2.2	**populateSubscriber()**	This method will receive an array as a parameter that will consist of the number of subscribers. The method must then populate the **subscribers** array with random numbers from 100000 to 500000 using the **Random class**.	4
5.2.3	**populateRatings()**	This method will receive two arrays as parameters that will consist of the user ratings as well as the service provider details. The method must then prompt the user to enter the ratings for each service provider (1-5), and then it must populate the **userRatings** array (**See Figure 5.1**).	4
5.2.4	**mostWatchedService()**	This method will receive an array as a parameter that will consist of the number of subscribers. This method must determine which service providers have the most subscribers. The method must then return the position number at which the most subscribers were found. **DO NOT USE THE SORT TECHNIQUE!!**	8

5.2.5	**sortServiceProvider()**	This method will receive four arrays as parameters that will consist of the **service provider**, the **country of origin**, the **number of subscribers**, and the **user's ratings** respectively. This method must then sort the arrays in ascending order using **service provider names**. Use the Bubble Sort technique.	12
5.2.6	**displayDetails()**	This method will receive four arrays as parameters that will consist of the **service provider**, the **country of origin**, the **number of subscribers**, and the **user's ratings** respectively. The method must then display the service provider details as indicated in **Figure 5.3**.	4

Sample Output

```
Enter user ratings(1-5) for YouTube: 4
Enter user ratings(1-5) for Netflix: 5
Enter user ratings(1-5) for DSTV: 3
Enter user ratings(1-5) for Showmax: 3
Enter user ratings(1-5) for PrimeTV: 2
Enter user ratings(1-5) for SABC+: 1
```

Figure 5.1 Prompt for number of subscribers

```
Enter user ratings(1-5) for YouTube: 4
Enter user ratings(1-5) for Netflix: 5
Enter user ratings(1-5) for DSTV: 3
Enter user ratings(1-5) for Showmax: 3
Enter user ratings(1-5) for PrimeTV: 2
Enter user ratings(1-5) for SABC+: 1

Most watched service provider YouTube 534240 subcribers
```

Figure 5.2 Display most watched service provider

```
Enter user ratings(1-5) for YouTube: 4
Enter user ratings(1-5) for Netflix: 5
Enter user ratings(1-5) for DSTV: 3
Enter user ratings(1-5) for Showmax: 3
Enter user ratings(1-5) for PrimeTV: 2
Enter user ratings(1-5) for SABC+: 1

Most watched service provider YouTube 534240 subcribers
================================================================================
Service Providers       Country          Subscribers     User Ratings
DSTV                    South Africa     308237           3
Netflix                 United States    511914           5
PrimeTV                 United States    514997           2
SABC+                   South Africa     207765           1
Showmax                 South Africa     511173           3
YouTube                 United States    534240           4
```

Figure 5.3 Display the sorted service providers

TOTAL FOR SECTION A (Question 1 - 3): 21

TOTAL FOR SECTION B (Question 4 - 5): 59

FULL MARKS: 90